Total No. of Questions : 8]

SEAT No. :

PC2819

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S.E. (Computer Engg.)/(Computer Science & Design Engg.)/(AI & DS)

COMPUTER GRAPHICS

(2019 Pattern) (Semester - III) (210244)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Figures to the right indicate full marks.***
3) ***Draw neat diagram wherever necessary.***
4) ***Assume suitable data, if necessary.***

Q1) a) Illustrate three types of Orthographic projections with suitable diagram. **[6]**

b) Represent the matrices used for following 3D transformations - Translation, Scaling, Shearing. **[6]**

c) Why homogeneous coordinates are used in 2D transformations. Illustrate 2D rotation about an arbitrary point with diagram. **[5]**

OR

Q2) a) Given a triangle with coordinates P (0, 0), Q (1, 0) and R(1, 1). Apply shear parameter 3 along Y axis and find out the new coordinates of the object. Draw the diagram. **[6]**

b) Illustrate three types of perspective projections with neat diagrams. **[6]**

c) In 3-D transformations, illustrate 3D object rotation about an arbitrary axis. **[5]**

Q3) a) Define the following terms with example: **[6]**

i) RGB Colour Model

ii) Diffuse reflection

iii) Specular reflection

b) Compare Halftone shading, Gauraud shading & Phong shading. **[6]**

c) Illustrate "Area Subdivision Warnock algorithm" with suitable example. **[5]**

OR

P.T.O.

Q4) a) Compare RGB, HSV & CMY colour models. **[6]**

b) Illustrate the Z-buffer algorithm in detail with diagram. **[6]**

c) How is back face detection and removal possible? Illustrate using algorithm. **[5]**

Q5) a) List various methods for drawing curved lines. Write short note on Bezier curve. **[6]**

b) Define fractal and give any two examples of fractal. **[7]**

c) Compare Bezier and B-spline curves. **[5]**

OR

Q6) a) State the properties of Bezier curve. Derive blending function of Bezier curve. **[6]**

b) How is Koch curve generated recursively? Elaborate the fractal dimension used in Koch curve. **[7]**

c) In curve drawing, define the terms control points, order of continuity, blending function for cubic polynomial. **[5]**

Q7) a) State the applications of Computer Graphics in Gaming Industry, what is morphing. **[6]**

b) State twelve animation principles that are applied while creation of sequences in Animation. **[6]**

c) Illustrate how segments in the display file are created, renamed, and deleted. **[6]**

OR

Q8) a) Explain renaming of a segment with suitable example. **[7]**

b) Write any three important features of NVIDIA gaming platform. **[4]**

c) Write a short note on motion specification methods based on: **[7]**

i) Geometric and kinematics information.

ii) Specification methods based on physical information.

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